

Prabudhd Krishna Kandpal

Machine Learning | Multimodal Systems | LLM Post-Training and Robustness

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EDUCATION

University of Southern California, Viterbi School of Engineering

M.S. Computer Science, Artificial Intelligence Track, GPA: 4.00

Los Angeles, CA

Aug 2025 - May 2027

Coursework: Deep Learning, Applied NLP, Machine Learning, Multimodal Learning, Algorithms, Artificial Intelligence

Guru Gobind Singh Indraprastha University

B.S. Artificial Intelligence and Machine Learning, GPA: 9.45/10, Bronze Medalist

Delhi, India

Nov 2021 - May 2025

INDUSTRY EXPERIENCE

Novartis

Business Planning and Analysis Intern - AI and Automation

East Hanover, NJ

Jun 2026 - Aug 2026

- Built a **LightGBM** adoption-propensity model from early diagnostic-testing and market signals for a new cardiovascular category, enabling field teams to rank likely early adopters and prioritize outreach.
- Developed conversational reporting agents with **AWS Bedrock** and retrieval-augmented generation. Added a deterministic calculation layer for auditable figures and reduced weekly reporting from hours to minutes.

ZS Associates - Pfizer

Decision Analytics Associate Intern

Delhi, India

Feb 2025 - Jul 2025

- Developed an **XGBoost** targeting model with **SHAP** explanations to rank early adopters while making feature attribution usable by commercial stakeholders.
- Formulated resource-allocation models across **16 portfolios**, balancing profitability, budget, and workload constraints.
- Executed A/B testing, ROI modeling, and coverage-gap clustering across Japan, Australia, and New Zealand to optimize promotional investment.
- Delivered Excel and Tableau analytics adopted by global teams, earning the **Best Case Study Solution** award and a full-time return offer.

LunexAI

AI/ML Intern

Delhi, India

Aug 2023 - Nov 2023

- Co-developed a GPT-3 and **ChromaDB** content-generation system, improving vector-storage and preprocessing workflows for real-time personalized retrieval in a pay-per-use SaaS product.

TEACHING EXPERIENCE

USC - Course Producer / Grader, CSCI 561: Foundations of Artificial Intelligence

Jan 2026 - May 2026

- Led office hours and mentored graduate students in search, game playing, logic, and probabilistic reasoning. Graded exams and programming assignments.

SELECTED PROJECTS

Most-Forgotten Replay for Continual Preference Tuning

CSCI 544 - In Progress

- Studying **catastrophic forgetting**: when an LLM is tuned sequentially for helpfulness, safety, and response quality, does it lose earlier behaviors, and can a fixed 10% replay budget preserve them?
- Implemented DPO with 4-bit **QLoRA** on Qwen2.5-1.5B-Instruct and ranked replay pairs by their drop in preference margin. The first matched validation cell improved mean retention by **0.75 points** and final quality by **2.5 points** over equal-budget random replay. The full order/seed grid is in progress.

Poisoned Paperwork: Adversarial Document VLMs

CSCI 566 - In Progress

- Testing whether visually subtle pixel changes can force a document VLM to extract a chosen false receipt total or resume credential. Implementing targeted **PGD/EOT** attacks against **Qwen2.5-VL-3B-Instruct**.
- Using **Donut** as a reproduction baseline and **InternVL2.5-4B** as a black-box transfer target across SROIE, CORD v2, and synthetic resumes. Measuring JPEG/resizing/printing survival, perceptibility, and adaptive defenses.

Predicting Emotional Descent in Dyadic Interaction

CSCI 535 - Multimodal Learning

- Built a multimodal transformer from scratch to locate the exact utterance where a calm conversation turns into conflict, aligning asynchronous speech prosody, facial affect, and text over **20+ hours** of dialogue. Reached **86.7% accuracy**, **7.2 points** above a Gemini API baseline.

PUBLICATIONS

Kandpal, P. K. et al. "Text Summarization and Sentiment Analysis Using Transformer-Based Models." Springer LNEE 1278, 2025.

Kandpal, P. K. et al. "Time Series Forecasting of NSE Stocks Using ARIMA, Prophet, and Stacked LSTM." Springer LNNS 788, 2023.

TECHNICAL SKILLS

ML: PyTorch, Transformers, DPO, QLoRA/PEFT, Multimodal ML, RAG, Adversarial ML, LightGBM, XGBoost, SHAP, Time Series, A/B tests

Languages/Data: Python, SQL, C/C++, R, Pandas, NumPy, Snowflake **Tools:** AWS Bedrock, Databricks, Dataiku, ChromaDB, Tableau, Git